

Mannich Reaction of Secondary Benzylzinc Reagents: Synthesis of α,β -Disubstituted β -Arylethylamines

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The organometallic multicomponent Mannich reaction of secondary benzylzinc compounds is described. These organometallic reagents were efficiently prepared by direct metalation of (1-bromoethyl)benzenes in tetrahydrofuran with zinc dust. Their subsequent multicomponent Mannich coupling with amines and aldehydes allowed the straightforward preparation of a large

variety of α,β -disubstituted β -arylethylamines in yields ranging from 13% to 79%. This reaction was found to require the use of a heated mixture of tetrahydrofuran and acetonitrile, tetrahydrofuran being essential to reach good yields and acetonitrile ensuring reproducibility.

1. Introduction

The β -arylethylamine scaffold is one of the most prevalent structural motifs found in molecules exhibiting biological activity (Scheme 1A). Their importance in drug discovery, conjugated with the need for new compounds, has led to the development of original and efficient chemical methods for their synthesis.^[1–4] However, whereas numerous strategies have been elaborated for the preparation of α -substituted amines, only a limited number of reactions allowed the direct synthesis of α,β -disubstituted β -arylethylamines. For instance, the preparation of β -methylphenylalanine derivatives has been achieved through the hydrogenation of the parent enamine.^[5–7] In order to circumvent the mandatory presence of an electron withdrawing group on the olefin, alternative routes have been occasionally used such as the addition of Me_2CuLi to an aziridine,^[8] or the condensation of a laterally lithiated *o*-ethyl benzamide with an imine (Scheme 1B).^[9] Nonetheless, despite the presence of this pattern in biologically active molecules such as corydaline, UH-232 or vafidemstat, there is no general strategy for the preparation of α,β -disubstituted β -arylethylamines, and the implementation of a straightforward method to access this structural motif should propel advancements in drug discovery.

In this context, the well-known Mannich reaction^[10] is undoubtedly one of the most efficient tools for the multicomponent synthesis of densely substituted amines. In the original version of the reaction, β -aminocarbonyl compounds are obtained by three-component coupling of enol/enolates, amines and carbonyl compounds. However, the quick structural and molecular diversification accessible by simple variation of one reaction component has prompted the organic chemist's community to develop modern versions of the reaction.^[11] Consequently, the synthesis of α -branched amines by multicomponent procedures derived from the Mannich reaction has been the subject of a tremendous development over the last years. This is particularly the case of reactions employing organometallic nucleophiles instead of enols/enolates of the original version.^[12] Among the diverse organometallic species having appropriate selective reactivity to enable the development of multicomponent processes related to the Mannich reaction, organozinc compounds are of particular interest.^[13–19] Fan and coworkers described the preparation of α -substituted β -arylethylamines in THF by an organometallic Mannich reaction involving the *in situ* metalation of benzyl bromide by Zn.^[20] Our group also described a related Barbier-like process in MeCN and demonstrated that the Mannich reaction is accelerated in this solvent.^[21,22] However, in both cases, only primary benzylic bromides were employed, which can be ascribed to the lack of commercially available secondary alkyl bromides as well as the major formation of Wurtz coupling originating by-products.^[23] Therefore, whereas heteroleptic primary benzylzinc reagents represent reliable nucleophiles, whether generated *in situ* or preformed,^[24,25] there is still a lack of a comprehensive study addressing the preparation and reactivity of secondary benzylic organozinc reagents in the organometallic Mannich reaction. This is the object of the present manuscript (Scheme 1C).

2. Results and Discussion

The metalation of (1-bromoethyl)benzene **1a**, one of the rare commercially available secondary benzylic bromides, was

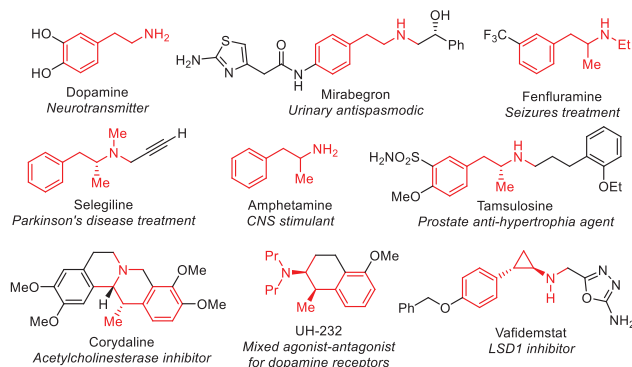
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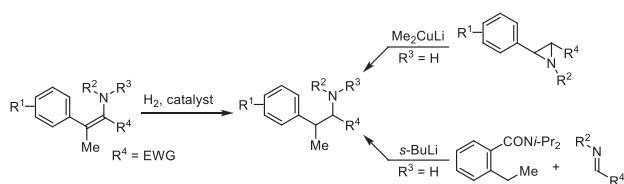
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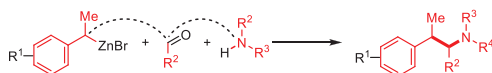
A. Selected examples of bioactive molecules incorporating the β -arylethylamine scaffold



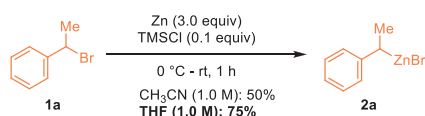
B. Main routes to α,β -disubstituted β -arylethylamines



C. Present work: multicomponent approach to the synthesis of α,β -disubstituted β -arylethylamines



Scheme 1. Importance of β -arylethylamines and synthetic strategies for their preparation.



Scheme 2. Zincation of (1-bromoethyl)benzene **1a**.

first envisaged. Building on our previous experience in the preparation of organozinc reagents, zincation was achieved using an excess of zinc dust (3 equivalents), pre-activated by chlorotrimethylsilane (TMSCl). However, in CH_3CN , which we have generally found to be the best-suited solvent, the reaction proved sluggish, leading to (1-phenylethyl)zinc bromide (**2a**) in only a moderate 50% yield. We quickly realized that THF was here a better solvent, providing organozinc **2a** in an improved 75% yield (Scheme 2). It was also found that, in order to obtain good yields, the benzyl bromide needs to be added dropwise over 30 min into the reaction medium cooled at 0°C , followed by an additional stirring at room temperature for a further 30 min.^[26]

Organozinc reagent **2a** thus obtained (3.0 equiv) was next engaged in a model Mannich reaction with benzaldehyde **3a** (1.2 equiv) and piperidine **4a** under various conditions. Results are reported in Table 1.

We first considered the use of THF as the sole solvent and after 16 h of reaction at 80°C , 60% of product **5aaa** was isolated in a 50/50 diastereomeric ratio (Table 1, entry 1). Although this result clearly demonstrates the possibility of carrying out the multicomponent coupling in THF alone, reactions were not so reproducible with a noticeable influence of the reaction concen-

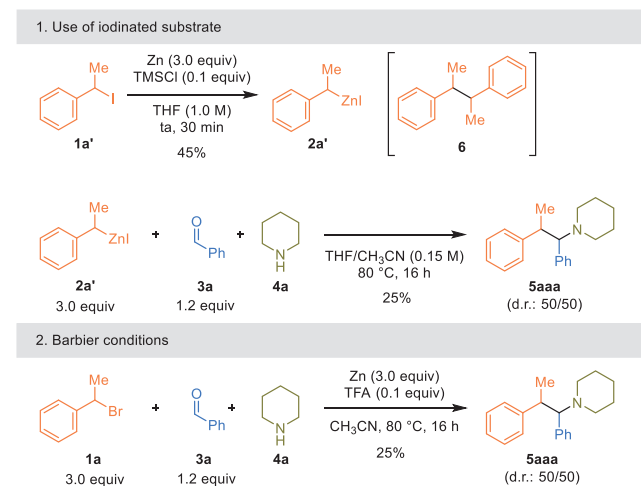
Table 1. Optimization of the reaction conditions.^{a)}

Entry	Additive	x (equiv)	Solvent(s)	T ($^\circ\text{C}$)	Yield (%) ^{b)}
1	–	–	THF	80	60
2 ^{c)}	–	–	CH_3CN	80	47
3	–	–	$\text{THF}/\text{CH}_3\text{CN}^{\text{d)}$	80	60
4 ^{e)}	–	–	$\text{THF}/\text{CH}_3\text{CN}^{\text{d)}$	80	32
5 ^{f)}	–	–	$\text{THF}/\text{CH}_3\text{CN}^{\text{d)}$	20	47
6	–	–	$\text{THF}/\text{CH}_3\text{CN}^{\text{d)}$	20	20
7	LiCl	1.0	$\text{THF}/\text{CH}_3\text{CN}^{\text{d)}$	80	38
8	CuCN	1.0	$\text{THF}/\text{CH}_3\text{CN}^{\text{d)}$	80	55

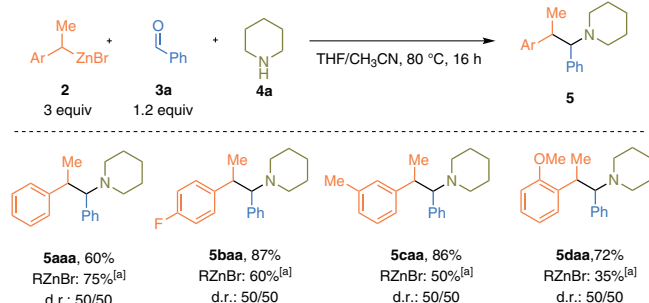
a) Reaction conditions: **2a** (3.0 equiv), **3a** (1.2 equiv), **4a** (0.5 mmol), 80°C (sealed tube), 16 h. b) Isolated yield. c) Metalation performed in CH_3CN . d) Mixtures were prepared by addition of the THF solution containing organozinc to MeCN. e) Dropwise addition of the organozinc compound. f) Reaction performed with 2.0 equiv of **2a**.

tration on yield. The use of MeCN was alternatively evaluated and delivered more robust results, although a lower average yield of 47% was obtained (Table 1, entry 2). We thus considered that most reliable reaction conditions may be achievable using mixtures of solvents and decided to investigate the use of THF/ CH_3CN mixtures, THF originating from the metalation step. This proved beneficial to the reaction outcome as a yield of 60% was again obtained with only a limited influence of the reaction concentration on yield (Table 1, entry 3). Several modifications were next assessed to improve the reaction yield. For instance, a dropwise addition of the organozinc reagent to a heated mixture of **3a** and **4a** as well as a decreased amount of **2a** led to only 32% and 47% of the reaction product, respectively (Table 1, entries 4 and 5). Moreover, when the reaction was carried out at room temperature, **5aaa** was isolated in a low 20% yield (Table 1, entry 6). On the other hand, the addition of LiCl had a deleterious effect, giving only 38% yield (Table 1, entry 7) and the use of a copper salt did not improve the reaction efficiency (Table 1, entry 8).

In order to investigate a possible influence of the halide substituent on the zinc atom of the benzylzinc reagent, the model Mannich reaction was next evaluated with **2a'** having a zinc iodide unit (Scheme 3, part 1). Using the conditions described above, compound **2a'** could be prepared in 45% yield. However, the reaction time had to be decreased to 30 min at room temperature to limit the amount of the Wurtz product **6** concomitantly formed. In the following three-component coupling, **2a'** also led to the formation of the product **5aaa** but in only 25% yield after 16 h at 80°C , showing the deleterious effect of switching to the iodo congener of **2a**. Moreover, in an attempt to improve the atom economy of the protocol, the reaction was also carried out under Barbier conditions (Scheme 3, part 2).^[20–22] The multicomponent coupling of bromide **1a**, benzalde-



Scheme 3. Modification of the pronucleophilic species.

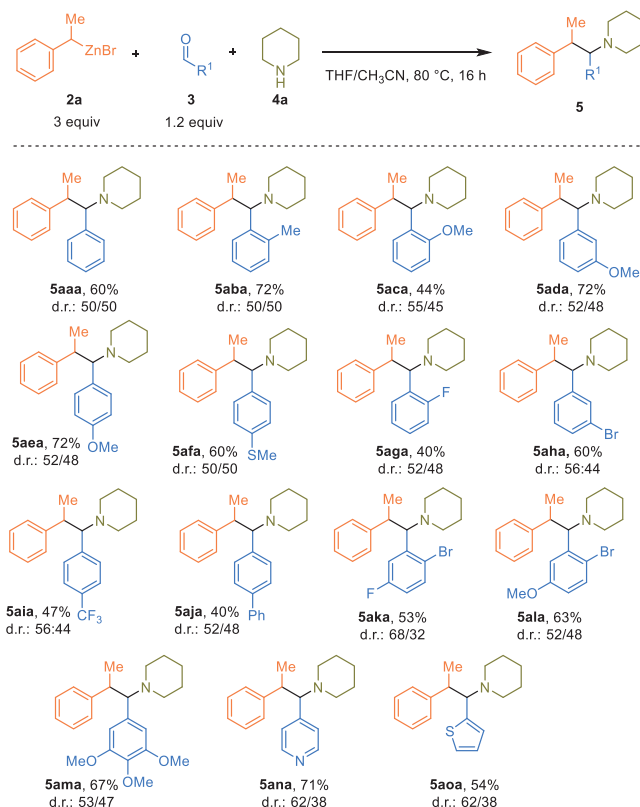


Scheme 4. Scope of secondary organozinc compounds. Reactions conditions: **2** (3.0 equiv), **3a** (1.2 equiv), **4a** (1 equiv), 80 °C (sealed tube), 16 h, THF/CH₃CN, 80 °C, 16 h. [a] Organozinc yield, determined by NMR after iodolysis.

hyde **3a** and piperidine **4a** was performed in the presence of 3.0 equivalents of zinc and 0.1 equivalents of TFA in acetonitrile, furnishing 25% of product **5aaa** after 16 h at 80 °C. Hence, the use of secondary benzylic nucleophiles in the Mannich reaction requires the use of a preformed organozinc species rather than an in situ generated one.

With optimized conditions in hands (Table 1, entry 3), the organozinc scope of the reaction was first evaluated with commercial or easily accessible reagents (Scheme 4). These results confirmed the efficiency of the reaction conditions, high yields being obtained with functionalized benzylzinc reagents, regardless of the position of the substituent on the phenyl ring. In addition, these results also demonstrated that the organozinc formation step has little influence on the subsequent three-component coupling, comparable results being obtained despite important disparity between organozinc yields.

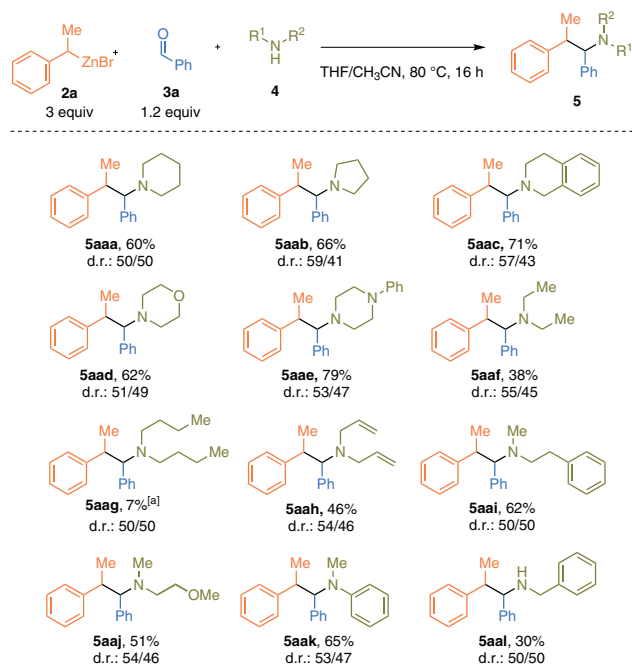
Aromatic aldehydes were next evaluated in the process (Scheme 5). These results show that the reaction tolerates both electron-donating (**5aca**: 44%; **5ada**: 72%; **5aea**: 72%) or electron-withdrawing (**5aga**: 40%; **5aha**: 60%; **5aia**: 47%) groups, regardless of the position. The steric hindrance caused by *ortho* substituents does not induce a notable decrease of the reaction efficiency (**5aba**: 72%). The reaction is also compatible with the presence of double and triple substitution on the



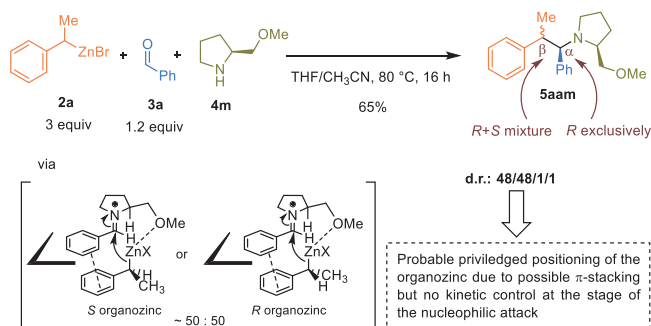
Scheme 5. Scope of aldehydes. Reactions conditions **2a** (3.0 equiv), **3** (1.2 equiv), **4a** (1 equiv), 80 °C, 16 h.

aldehyde, good yields being obtained in these cases (**5aka**, **5ala**, **5ama**, from 53% to 67% yield). Finally, heteroaromatic aldehydes were tested in the reaction, using pyridine- and thiophene based aldehydes, furnishing 71% (**5ana**) and 54% (**5aoa**) yield of the final product. It can be noted that reactions with aliphatic aldehydes were evaluated to complete the study, but no reaction was observed. Once again, the majority of diastereomeric ratios were close to 50/50, except for reactions employing heteroaromatic aldehydes, where ratios of 62/38 were obtained. The relative configuration of the major diastereomer could not be determined.

The amine scope of the reaction was finally studied (Scheme 6). Following the optimization part where piperidine showed significant reactivity, alicyclic amines were first tested in the reaction. Thus, whether with pyrrolidine, morpholine, 1-phenylpiperazine or even 1,2,3,4-tetrahydroisoquinoline, isolated yields between 62% and 79% were obtained (products **5aab**, **5aac**, **5aad** and **5aae**). Results observed with acyclic secondary amines showed less efficiency than previous Mannich reactions, with lower yields obtained. Thus, diethylamine, dibutylamine, and diallylamine led to 38%, 7%, and 46% of the products **5aaf**, **5aag** and **5aah** respectively, whereas the particularly low yield obtained with dibutylamine may be partly explained by steric issues. The introduction of a substituted aliphatic chain led to yields comparable to those observed with cyclic amines, 62% and 51% yields being obtained for products **5aai** and **5aj**, respectively. Finally, a secondary aniline derivative, *N*-methylaniline, also gave a good yield (**5aak**, 65%). The case



Scheme 6. Scope of amines. Reaction conditions 2a (3.0 equiv), 3a (1.2 equiv), 4 (1 equiv), 80 °C, 16 h. [a] GC yield with internal standard.



Scheme 7. Reaction involving a chiral amine derived from proline.

of 5aal is of particular interest. Indeed, a primary amine like benzylamine proved useable in the reaction without further activation, although only 30% of the product was isolated. It can also be noted that in all cases, diastereomeric ratios close to 50/50 are observed.

In order to gain more insight into the origin of this lack of diastereoselectivity, the optimized conditions were applied to a chiral amine derived from proline, (S)-2-(methoxymethyl)pyrrolidine 4m, which has been previously exploited in successful diastereoselective A³-coupling between aldehydes, alkynes and amines.^[27–30] Under the optimized conditions, product 5aam was obtained in 65% yield after 16 h at 80 °C (Scheme 7).^[31] From the four possibly formed diastereomers, only two were mainly detected by GC analysis in a ~1:1 ratio. No reaction was observed at 50 °C, precluding a possible increase of diastereoselectivity by lowering the temperature. Although it was not possible to determine accurately which isomers were majorly formed, one can reasonably consider that the reaction presents an important

stereofacial selectivity of the nucleophilic attack on the transient iminium, similarly to what has been previously described.^[27–30]

This would lead to the control of the configuration of the α -stereocenter. However, regarding the stereochemical outcome of the reaction, no kinetic selectivity of the attacking organozinc enantiomer was observed leading to an absence of control of the β -stereocenter. Therefore, it can be assumed that the two major diastereomers only differ from the configuration of the carbon atom β to the nitrogen and originating from the organozinc reagent.

3. Conclusion

In conclusion, we have shown that secondary benzylic organozinc reagents are convenient nucleophiles for the organometallic multicomponent Mannich reaction. These organozinc compounds can be efficiently prepared by treating the corresponding halide with zinc dust in THF. However, their use requires careful control of the solvent conditions. Therefore, by using a mixture of THF and acetonitrile, they proved to be valuable reaction partners in three-component reactions with a variety of aromatic aldehydes and amines. The scope of the reaction is broad, allowing for the preparation of a wide variety of α,β -disubstituted β -arylethylamines. Even if no simple diastereoselection was noted, preliminary results show that the α -stereocenter can be controlled by stereoinduction from a chiral auxiliary on the secondary amine.

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Conflict of Interests

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the [supplementary material](#) of this article.

Keywords: β -Arylethylamines · Amines · Aldehydes · Multicomponent reaction · Organozinc compounds

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